

Differential and Integral Calculus 1M. Schedule			
Week	Lesson	Contents	Stewart's book sections
1	1	Numbers, inequalities and absolute values. Coordinate geometry and lines.	Appendixes A and B
	2	Graphs of second-degree equations. Trigonometry.	Appendixes C and D
2	3	Four ways to represent a function. Mathematical models: A catalog of essential functions.	Sections 1.1-1.2
	4	New functions from old functions. Exponential functions. Inverse functions and logarithms.	Sections 1.3-1.5
3	5	The tangent and velocity problems. The limit of a function. Calculating limits using the limit laws.	Sections 2.1-2.3
	6	The precise definition of a limit.	Section 2.4
4	7	Continuity. Limits at infinity. Horizontal asymptotes.	Sections 2.5-2.6
	8	Derivatives and rates of change. The derivative as a function.	Sections 2.7-2.8
5	9	Derivatives of polynomials and exponential functions. The product and quotient rules. Derivatives of trigonometric functions.	Sections 3.1-3.3
	10	The chain rule. Implicit differentiation. Derivatives of logarithmic functions. Hyperbolic functions.	Sections 3.4-3.6, 3.11
6	11	Maximum and minimum values. The mean value theorem.	Sections 4.1-4.2
	12	How derivatives affect the shape of a graph. Indeterminate forms and l'Hôpital's rule.	Sections 4.3-4.4
7	13	Summary of curve sketching. Optimization problems.	Sections 4.5, 4.7
	14	Antiderivatives.	Sections 4.9
8	15	Areas and distances. The definite integral.	Sections 5.1-5.2
	16	The fundamental theorem of calculus. Indefinite integrals and the net change theorem. The substitution rule.	Sections 5.3-5.5
9	17	Areas between curves. Volumes. Volumes by cylindrical shells. Average value of a function.	Sections 6.1-6.3, 6.5
	18	Integration by parts. Trigonometric integrals. Trigonometric Substitution.	Sections 7.1-7.3
10	19	Integration of rational functions by partial fractions.	Section 7.4
	20	Strategy for integration. Improper integrals.	Sections 7.5, 7.8
11	21	Arc length. Area of a surface of revolution. Probability.	Sections 8.1-8.2, 8.5
	22	Sequences.	Sections 11.1
12	23	Series. The integral test and estimates of sums. The comparison tests.	Sections 11.2-11.4
	24	Alternating series. Absolute convergence and the ratio and root tests. Strategy for testing series.	Sections 11.5-11.7
13	25	Power series. Representations of functions as power series.	Sections 11.8-11.9
	26	Taylor and Maclaurin series. Applications of Taylor polynomials.	Sections 11.10-11.11

